# Task 6 — Electronics & Hardware

## Wiring diagram

- Mene jo bhi components use kia saare ek saaath koi online software me nahi the so ill give a circuit table and a rough hand drawn diagram ...

### 1. Main power path

| # | From | To | Notes |
|---|---|---|---|
| 1 | Battery + (XT60) | Fuse in | Fuse goes in the **positive** lead only: 3 A slow-blow, 32 V rated. |
| 2 | Fuse out | Power switch terminal 1 | Switch must be connected properly. (bohot baar ulta laga dia he mane) |
| 3 | Switch terminal 2 | yaha +ve of battery split into 2(say W1 and W2 | Splice that feeds both bucks. |
| 4 | W1 | Buck A (5 V) IN+ | |
| 5 | W2 | Buck B (6 V) IN+ | |
| 6 | Battery − | GND star point | The star point is Buck B's IN− |
| 7 | Battery JST-XH balance lead | Not connected | Use it only for charging(ofc) |

### 2. Buck converters

| # | From | To | Notes |
|---|---|---|---|
| 10 | Buck A OUT+ | +5V rail | Set to 5.00 V by using a multimeter and also no load |
| 11 | Buck A OUT− | GND of 5v rail | |
| 12 | Buck B OUT+ | +6V rail | Set to 6.00 V with no load first. |
| 13 | Buck B OUT− | GND of 6v rail | Short, thick wire, because this is the motor return path. |

### 3. Power to the ESP32 and QTR (5 V and 3V3)

| # | From | To | Notes |
|---|---|---|---|
| 14 | +5V rail | ESP32 VIN (or 5V) | |
| 15 | +5V rail | QTR-8RC VCC | Parallel with the ESP32. The QTR operates from 3.3–5 V. |
| 16 | GND of 5v rail | ESP32 GND | |
| 17 | GND of 5v rail | QTR-8RC GND | |
| 18 | ESP32 3V3 | TB6612 VCC | |

| | | | |
|---|---|---|---|
| 19 | ESP32 3V3 | TB6612 STBY | The TB6612FNG IC features built-in power management to conserve energy and prevent uncommanded motor behavior. So basically giving it 3.3 means the driver is always “on” |
| 20 | ESP32 3V3 | Left encoder VCC | We want to map the maze also na ….. |
| 21 | ESP32 3V3 | Right encoder VCC | |

## 4. Motor driver power and outputs

| # | From | To | Notes |
|---|---|---|---|
| 22 | +6V rail | TB6612 VM | Add 10 µF plus 0.1 µF at the pin, and optionally 100–470 µF bulk. |
| 23 | GND of 6v rail | TB6612 GND (both pins) | Connect both ground pins. |
| 24 | TB6612 AO1 | Left motor M1 | Swap M1 and M2 if the direction is wrong. |
| 25 | TB6612 AO2 | Left motor M2 | |
| 26 | TB6612 BO1 | Right motor M1 | |
| 27 | TB6612 BO2 | Right motor M2 | |

## 5. Signals: ESP32 GPIO map

- I picked pins that avoid the strapping pins (0, 2, 5, 12, 15) and the flash pins (6–11).
    - STRAPPING PINS (something i also learned now only!!)
        - Strapping pins" are special pins that the ESP32 reads their voltage level (HIGH or LOW) tells the internal processor how to start up
        - If you connect a motor driver input, sensor, or pull-up/pull-down resistor to one of these pins, that external component might force the pin HIGH or LOW during boot.
        - The ESP32 gets confused during startup and enters a "bootloop" or refuses to run your code at all until you unplug the pin.
    - Flash Pins
        - GPIO pins 6, 7, 8, 9, 10, and 11 are hardwired on the PCB directly between the ESP32 processor and that flash chip. They constantly transfer data back and forth while the robot runs.
        - If you try to use GPIO 6–11 as regular inputs or outputs for your motor driver, sensors, or encoders, you break the connection between the CPU and its memory.
        - The ESP32 instantly crashes or freezes, because it can no longer read the instructions of your program
- So avaoiding all the above

| Device pin | ESP32 GPIO | Notes |
|---|---|---|
| TB6612 PWMA | 25 | Left motor speed |
| TB6612 AIN1 | 26 | Left motor direction |
| TB6612 AIN2 | 27 | Left motor direction |
| TB6612 PWMB | 23 | Right motor speed |

| | | |
|---|---|---|
| TB6612 BIN1 | 18 | Right motor direction |
| TB6612 BIN2 | 19 | Right motor direction |
| Left encoder C1 / C2 | 34 / 35 | Input-only pins, which suit encoders. They have no internal pull-ups. |
| Right encoder C1 / C2 | 36 / 39 | Same as above. |
| Left and right encoder GND | GND star | Keep this separate from the motor's M2 pin. |
| QTR-8RC OUT1–OUT8 | 4, 13, 14, 16, 17, 21, 22, 32 | One GPIO each. Read them as timed pulses. |
| QTR-8RC LEDON | 33 | The LEDs are on by default, and pulling this pin low turns them off. You can leave it unconnected. |

## Troubelshooting

- Voltage drop in the power supply
    - When the motors start, they can draw a large startup current. If the battery, wiring or buck converter cannot supply that current, the voltage may drop enough to reset the ESP32
    - Solving it -
        - Observe the 5 V rail during startup using a multimeter, if the 5 V rail drops sharply at the moment of reset, investigate the battery voltage, buck converter's current capability, wiring and connections first (ye mere saath hua tha utkarsh me .. wo car mera comptetion ke pehle tak sab sahi chal rha tha comptetion me disconnect hona shuru kar dia .. i believe wo voltage drop ke vajah se hi ho rha tha and mene vo esp ke ke vin and gnd pe capicator laga ke try kia but still ho rha tha .. still dont know ke wo kaise fix karna 🥀 🥀 🥀 ....)

- Poor grounding or loose connections
    - Ye esily fixable he you jus need to see ki koi connctions loose toh anhi he and thats it .. very common but also very easily fixable